$FINA\ 6339 : Lecture\ 7\ \&\ Lecture\ 8.$

[Portfolio Optimization (Part I)] [R].)

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Spring 2026.

```r
# FINA 6339: Lecture 7 & Lecture 8
# [Portfolio Optimization: Part I]
##########################################################

#######################################
# Home Practice (Problem 1)
# Portfolio Optimization (Blue Chips).
#######################################

####################
# Data Description:
####################

## Daily simple and log returns for Nike, Amazon,
## Google, Tesla, and Microsoft, covering the period
# Jan 2020 - December 2022.

## The data set will be posted on Canvas as
## FiveStocksSimpleRet.csv and FiveStocksLogRet.csv file.

##########################################################

# Define the working directory
setwd("C:\\Users\\Milivoje Davidovic\\Desktop")

# Import data set into R environment
mydata = read.csv("FiveStocksLogRet.csv") # Log returns
attach(mydata)

# Create Date vector
Date=seq(as.Date("2020-01-01"), by=1, len=756)

# Plot Log Returns
windows(8,10)
plot(Date, NIKE, type="l", col="red",
     main="Blue Chips Log Returns (Jan 2020 - Dec 2022)",
     ylab="Return",ylim=c(-0.06,0.06))
```

```r
lines(Date, AMZN, col="blue", lty=2)
lines(Date, GOOGLE, col="green", lty=2)
lines(Date, TESLA, col="orange", lty=2)
lines(Date, MSFT, col="black", lty=2)
grid(nx=20,ny=20)
# Add a legend
legend("topleft", legend=c("NIKE", "AMZN","GOOGLE","TESLA","MSFT"),
       col=c("red", "blue","green","orange","black"),
       lty=1, cex=0.8)
```

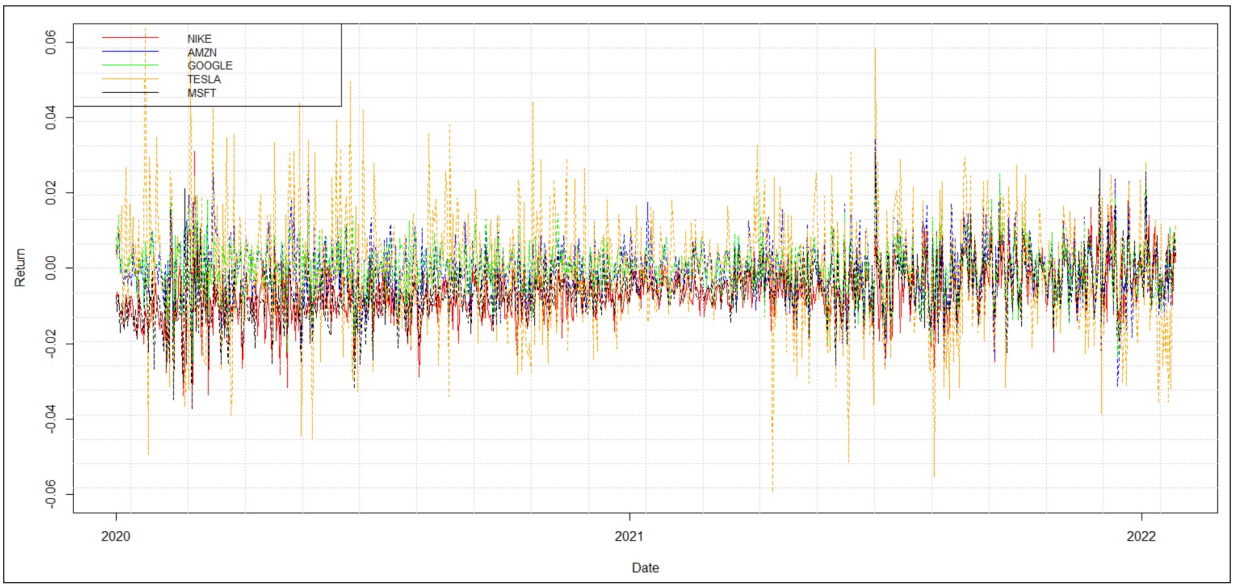

Figure 1: Log Returns (Blue Chips).

```r
##################################################
# Portfolio Optimization (Variance Minimization).
##################################################
returns=ts(mydata[,2:6]) # to cut of the Date
# ts: create time series vector
print(head(returns, 5))
```

```
##              NIKE         AMZN       GOOGLE       TESLA         MSFT
## [1,] -0.008429582  0.005297210  0.008279190 0.005853294 -0.007248433
## [2,] -0.006309223  0.002431931  0.004104826 0.002467612 -0.011430121
## [3,] -0.008244359  0.009898444  0.013994444 0.010779879 -0.006894031
## [4,] -0.012141780  0.000537850 -0.001431445 0.007150800 -0.017021422
## [5,] -0.010986244 -0.001391101  0.003801931 0.016585289 -0.009093842
```

```r
# Get a character vector of the fund names
funds <- colnames(returns)

# Install/call the required libraries
suppressMessages(library(PortfolioAnalytics))
```

```r
suppressMessages(library(foreach))
suppressMessages(library(iterators))
```

```r
suppressMessages(library(ROI))
suppressMessages(library(ROI.plugin.quadprog))
suppressMessages(library(ROI.plugin.glpk))

# Create portfolio object
MinVar <- portfolio.spec(assets=funds)

# Add full investment constraint to the portfolio object
MinVar <- add.constraint(portfolio=MinVar, type="full_investment")

# Add objective to minimize variance
MinVar <- add.objective(portfolio=MinVar, type="risk", name="var")

# Run the optimization procedure (Global Min variance)
opt_gmv <- optimize.portfolio(R=returns, portfolio=MinVar,
          optimize_method="ROI", trace=TRUE)

# OPTIMAL WEIGHTS
print(opt_gmv)
```

```
## ***********************************
## PortfolioAnalytics Optimization
## ***********************************
##
## Call:
## optimize.portfolio(R = returns, portfolio = MinVar, optimize_method = "ROI",
##     trace = TRUE)
##
## Optimal Weights:
##    NIKE    AMZN  GOOGLE   TESLA    MSFT
##  0.3445  0.0989  0.4917 -0.0214  0.0863
##
## Objective Measure:
##   StdDev
## 0.006303
```

```r
# Objective: StdDev = 0.006303

##########################################
# Visualization of the Optimal portfolio.
##########################################
windows(8,10)
plot(opt_gmv, chart.assets=TRUE)
```

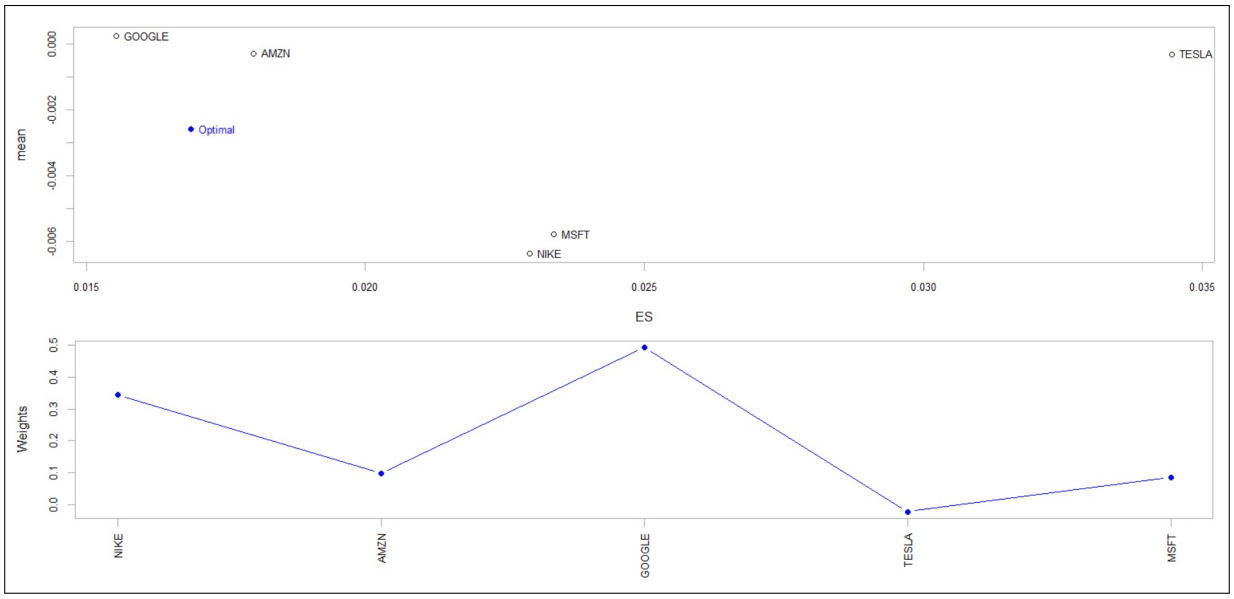

Figure 2: Optimal Portfolio (Min-Var).

```
############################
# Min-Var Optimal Portfolio.
############################

# Create the optimal portfolio using these weights
returns=mydata[,2:6]
OptPort1 = (0.3445*returns$NIKE) + (0.0989*returns$AMZN)+
  (0.4917*returns$GOOGLE)+(-0.0214*returns$TESLA)+
  (0.0863*returns$MSFT)

# Distribution of your optimal portfolio
windows(8,10)
hist(OptPort1, main="Optimal Min-Var Portfolio Retruns",
     xlab="Returns",col="orange",breaks = 30)
```

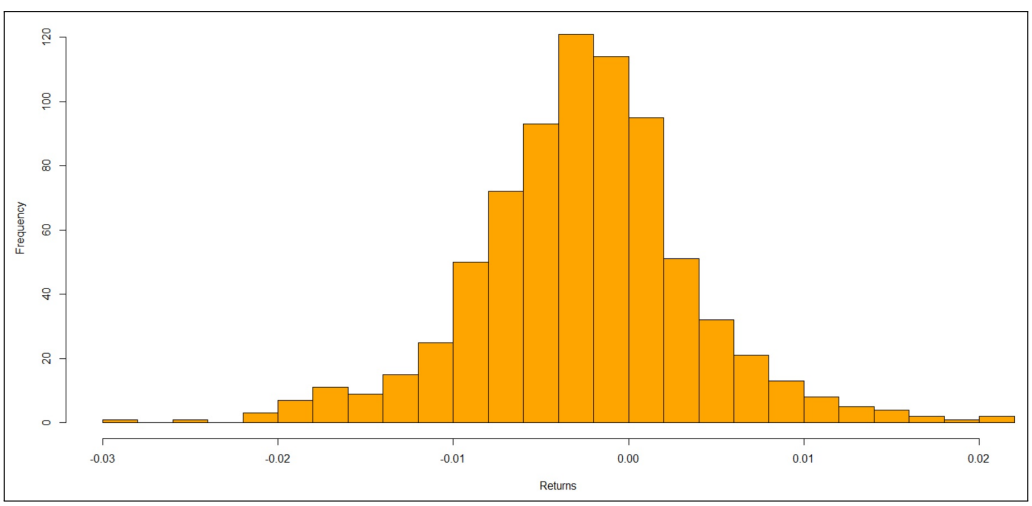

Figure 3: Optimal Min-Var Portfolio Returns.

```r
########################################
# Maximizing Mean Return (Liner Utility)
########################################

# Create portfolio object
MaxRet <- portfolio.spec(assets=funds)

# Add constraints to the portfolio object

## Full investment constraint
MaxRet <- add.constraint(portfolio=MaxRet,
                         type="full_investment")

## Add objective to the portfolio object
MaxRet <- add.objective(portfolio=MaxRet,
type="return", name = "mean")

## Note: Box Constraint is necessary for MaxRet problem
## Otherwise, the weights would go from -inf to inf
MaxRet <- add.constraint(portfolio=MaxRet, type="box",
                         min=c(0, 0, 0, 0,0),
                         max=c(0.25,0.25,0.25,0.25,0.25))

## To see the summary of your portfolio specification
print(MaxRet)
```

```
## ***************************************************
## PortfolioAnalytics Portfolio Specification
## ***************************************************
##
## Call:
## portfolio.spec(assets = funds)
##
## Number of assets: 5
## Asset Names
## [1] "NIKE"   "AMZN"   "GOOGLE" "TESLA"  "MSFT"
##
## Constraints
## Enabled constraint types
##      - full_investment
##      - box
##
## Objectives:
## Enabled objective names
##      - mean
```

```r
# Run the optimization
returns=ts(mydata[,2:6]) # returns must be a ts object
opt_max_ret <- optimize.portfolio(R=returns, portfolio=MaxRet,
          optimize_method="ROI", trace=TRUE)

# Optimal Weights
print(opt_max_ret)
```

```
## **********************************
```

```
## PortfolioAnalytics Optimization
## **********************************
##
## Call:
## optimize.portfolio(R = returns, portfolio = MaxRet, optimize_method = "ROI",
##     trace = TRUE)
##
## Optimal Weights:
##   NIKE   AMZN GOOGLE  TESLA   MSFT
##   0.00   0.25   0.25   0.25   0.25
##
## Objective Measure:
##      mean
## -0.001539
```

```
##########################################
# Visualization of the Optimal portfolio.
##########################################
windows(8,10)
plot(opt_max_ret, chart.assets=TRUE)
```

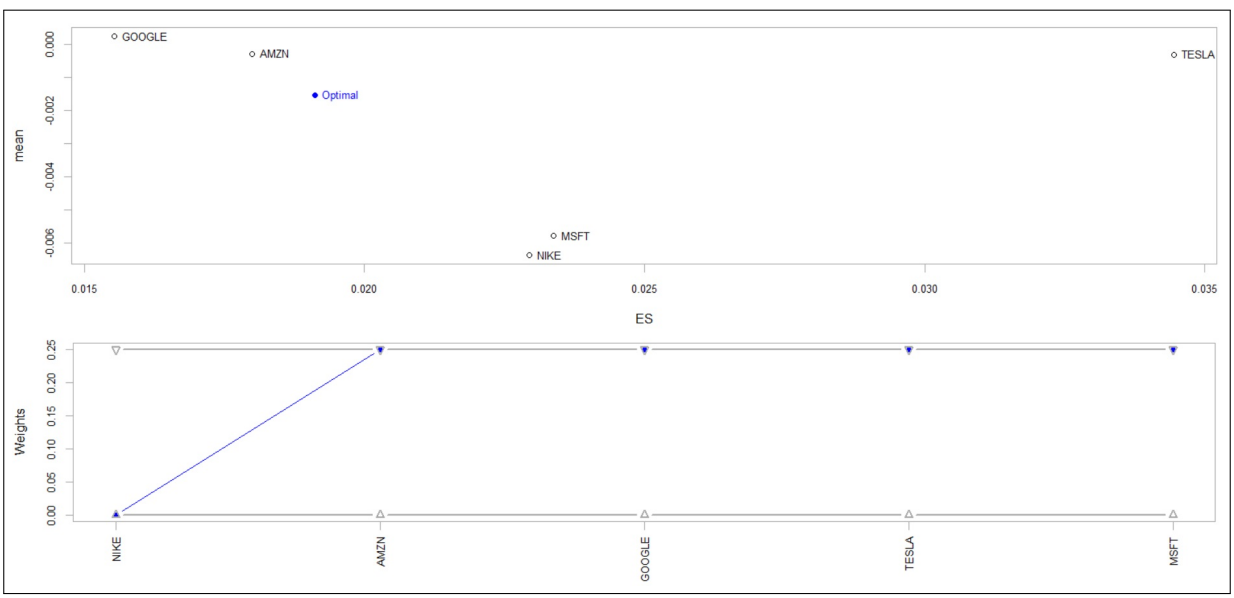

Figure 4: Optimal Portfolio (Max-Ret).

```
################################
# Max-Return Optimal Portfolio.
################################

# Create the optimal portfolio using these weights
returns=mydata[,2:6]
OptPort2 = (0.25*returns$NIKE) + (0.25*returns$AMZN)+
  (0.25*returns$GOOGLE)+(0.25*returns$TESLA)+
  (0.25*returns$MSFT)

# Distribution of your optimal portfolio
```

```r
windows(8,10)
hist(OptPort2, main="Optimal Max-Ret Portfolio Returns",
     xlab="Returns",col="green",breaks = 30)
```

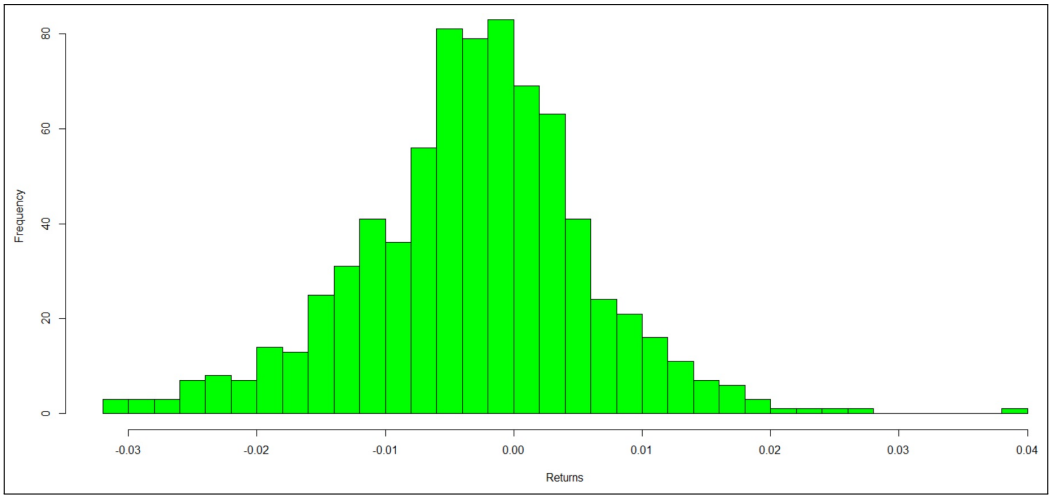

Figure 5: Optimal Max-Return Portfolio Returns.

```r
##################################
# Complex Portfolio Optimization.
##################################

# We can add different constraints to optimize
# portfolio weights using a random portfolio.

# Example: Our objective is to maximize portfolio
# return with a target of 0.002 and minimize portfolio
# StdDev with a target of 0.025 subject to dollar neutral (DN),
# box, and position limit constraints.

Port_DN<- portfolio.spec(assets=funds)

# Add constraint such that the portfolio weights sum to 0*
Port_DN <- add.constraint(Port_DN, type="weight_sum",
                          min_sum=-0.01, max_sum=0.01)

# Box Constraint: weight_i=(-0.25,0.25)
Port_DN <- add.constraint(Port_DN, type="box",
                          min=-0.25, max=0.25)

# Position Limit: at most 3 positions
Port_DN <- add.constraint(Port_DN, type="position_limit",
                          max_pos=3)

#######################
# Portfolio Objectives.
#######################
```

```r
# Goal 1: Maximize portfolio return with a target of 0.002;
Port_DN_SD <- add.objective(Port_DN, type="return",
                            name="mean", target=0.00015)

# Goal 2: Minimize portfolio StdDev with a target of 0.025
Port_DN_SD<- add.objective(Port_DN_SD, type="risk", name="StdDev",
                  target=0.0002)


# Generate random portfolios
RPort <- random_portfolios(Port_DN, 10000, "sample")

# Run the optimization
returns=ts(mydata[,2:6])
Opt_DN_Port <- optimize.portfolio(returns, Port_DN_SD,
                              optimize_method="random", rp=RPort,
                              trace=TRUE)
```

```
## Warning: executing %dopar% sequentially: no parallel backend registered
```

```r
## Optimal Weights
print(Opt_DN_Port)
```

```
## ***********************************
## PortfolioAnalytics Optimization
## ***********************************
##
## Call:
## optimize.portfolio(R = returns, portfolio = Port_DN_SD, optimize_method = "random",
##     trace = TRUE, rp = RPort)
##
## Optimal Weights:
##   NIKE   AMZN GOOGLE  TESLA   MSFT
## -0.012  0.032  0.000  0.000 -0.016
##
## Objective Measures:
##      mean
## 0.0001601
##
##
##    StdDev
## 0.0002084
```

```r
##############################################################
# Visualization of the Optimal Portfolio its "neighbours".
##############################################################

windows(8,10) # new plotting device
plot(Opt_DN_Port, main="Dollar Neutral Portfolio",
     risk.col="StdDev", neighbors=20)
```

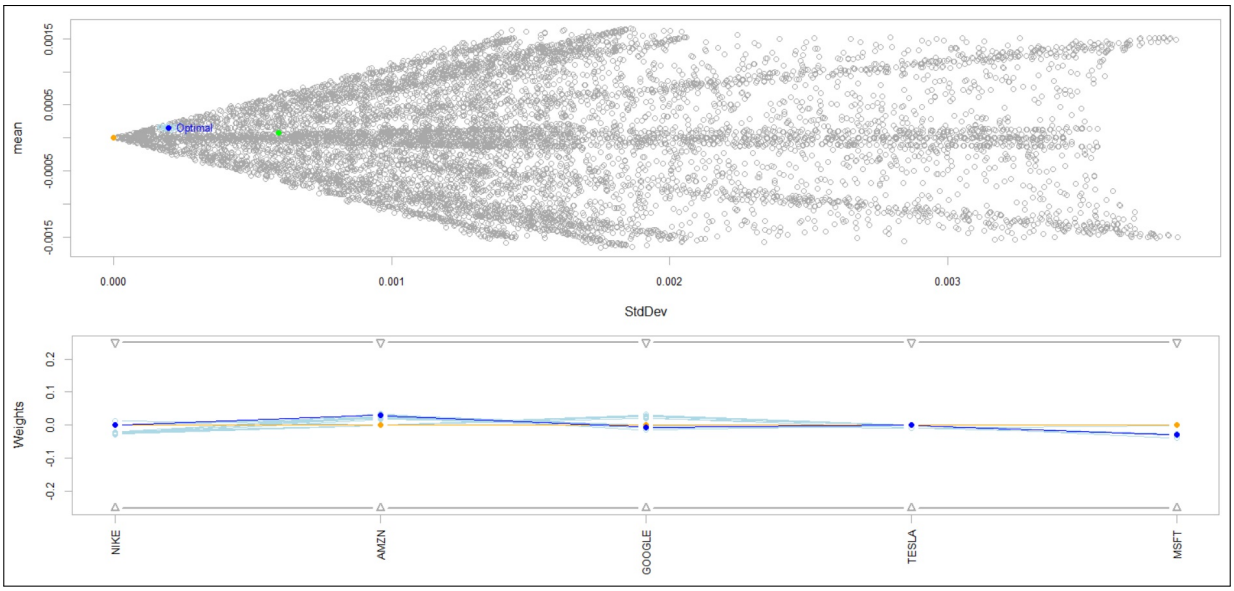

Figure 6: Dollar Neutral Portfolio.

```
# Calculate the optimal portfolio returns
OptPort3 = (0.3*AMZN) + (-0.26*MSFT)

# Distribution of your optimal portfolio
windows(8,10)
hist(OptPort3, main="Optimal Dollar-neutral
     Portfolio Returns",
     xlab="Returns",col="orange",breaks = 30)
```

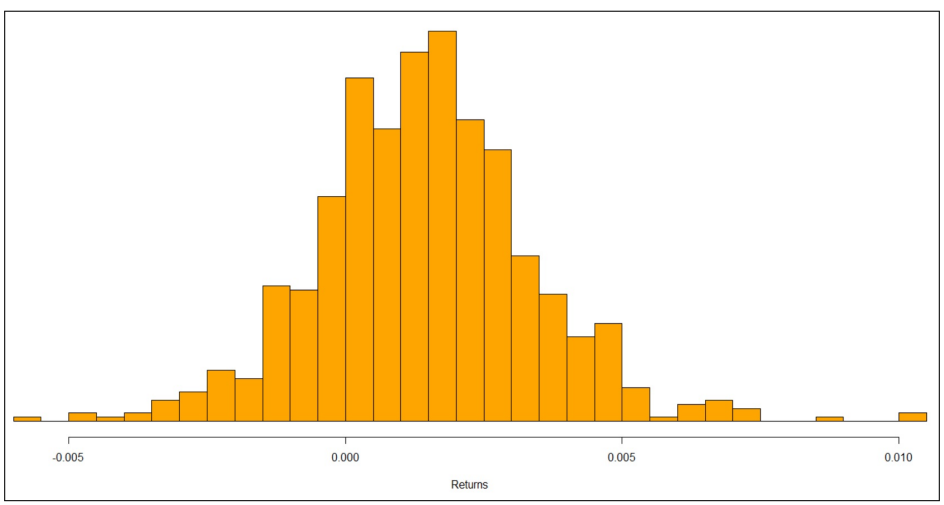

Figure 7: Dollar Neutral Portfolio Returns.

```
#######################################################
# Compare the Three Optimal Solutions.
# [Min-Var vs. Max-Ret vs. Dollar Neutral Allocations].
#######################################################
```

```r
data <- data.frame(Weights = c(0.3445, 0.0989, 0.4917, -0.0214, 0.0863,
                               0.00,0.25,0.25,0.25,0.25,
                               0.000,0.030,0.000,0.000,-0.026),  # Create example data
                   Optimal_Portfolios = rep(c("Min-Var",
                                              "Max-Ret",
                                              "Dollar-Neutral"),
                                            each = 5),
                   Companies = c("NIKE",
                                 "AMZN",
                                 "GOOGLE",
                                 "TESLA","MSFT"))
# Check you weights distribution
data
```

```
##    Weights Optimal_Portfolios Companies
## 1   0.3445            Min-Var      NIKE
## 2   0.0989            Min-Var      AMZN
## 3   0.4917            Min-Var    GOOGLE
## 4  -0.0214            Min-Var     TESLA
## 5   0.0863            Min-Var      MSFT
## 6   0.0000            Max-Ret      NIKE
## 7   0.2500            Max-Ret      AMZN
## 8   0.2500            Max-Ret    GOOGLE
## 9   0.2500            Max-Ret     TESLA
## 10  0.2500            Max-Ret      MSFT
## 11  0.0000     Dollar-Neutral      NIKE
## 12  0.0300     Dollar-Neutral      AMZN
## 13  0.0000     Dollar-Neutral    GOOGLE
## 14  0.0000     Dollar-Neutral     TESLA
## 15 -0.0260     Dollar-Neutral      MSFT
```

```r
# Plot using ggplot (Grouped barplot)
library(ggplot2)
windows(8,10)
ggplot(data,
       aes(x = Optimal_Portfolios,
           y = Weights,
           fill = Companies)) +
  geom_bar(stat = "identity",
           position = "dodge")+
  geom_text(aes(label= Weights),
            position=position_dodge(width=0.9),
            vjust=-0.25)
```

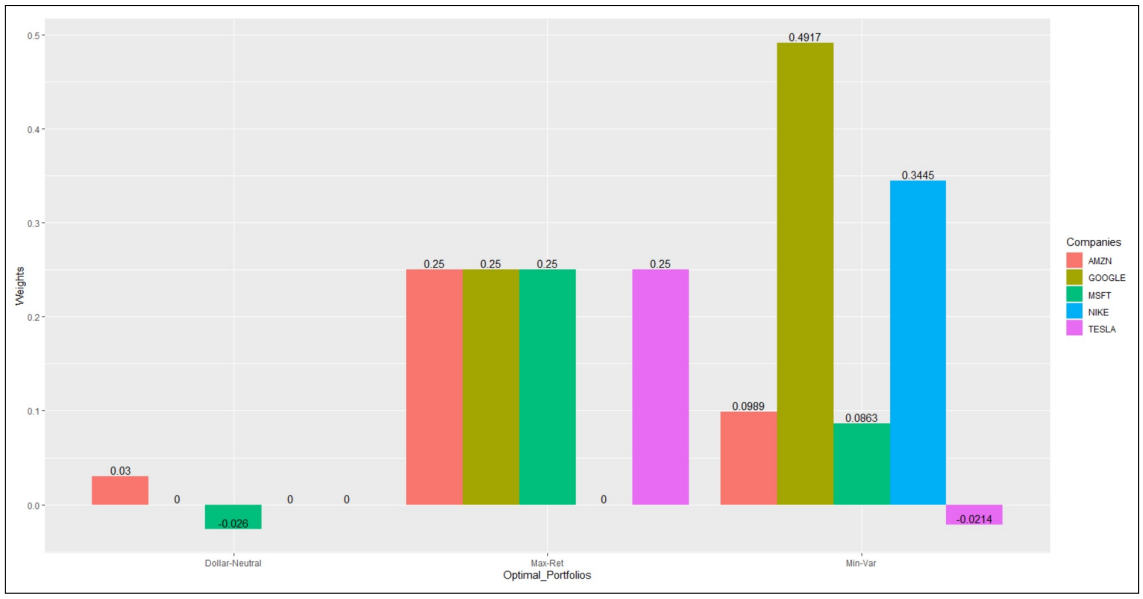

Figure 8: Optimal Portfolio Weights (Comparison).

```
##############################################
# Descriptive Statistics [Optimal Portfolios].
##############################################
library(pastecs)
```

```
##
## Attaching package: 'pastecs'

## The following objects are masked from 'package:xts':
##
##     first, last
```

```
MinVar = OptPort1
MaxRet = OptPort2
DollNeut = OptPort3

OptPortfolios = cbind(MinVar,MaxRet,DollNeut)
stat.desc(OptPortfolios)
```

```
##                     MinVar        MaxRet      DollNeut
## nbr.val       7.560000e+02  7.560000e+02  7.560000e+02
## nbr.null      0.000000e+00  0.000000e+00  0.000000e+00
## nbr.na        0.000000e+00  0.000000e+00  0.000000e+00
## min          -2.957777e-02 -3.197828e-02 -5.892492e-03
## max           2.054574e-02  3.946806e-02  1.036073e-02
## range         5.012351e-02  7.144635e-02  1.625322e-02
## sum          -1.967119e+00 -2.368528e+00  1.074760e+00
## median       -2.490672e-03 -2.773964e-03  1.358397e-03
## mean         -2.602009e-03 -3.132973e-03  1.421640e-03
## SE.mean       2.292295e-04  3.261137e-04  7.164505e-05
## CI.mean.0.95  4.500030e-04  6.401973e-04  1.406472e-04
## var           3.972491e-05  8.040069e-05  3.880558e-06
## std.dev       6.302770e-03  8.966643e-03  1.969913e-03
## coef.var     -2.422271e+00 -2.862024e+00  1.385663e+00
```

```r
#####################################
# Plotting Optimal Portfolio Returns.
#####################################
windows(8,10)
plot(Date, MinVar, type="l", col="red",
     main="Optimal Portfolio Log Returns (Jan 2020 - Dec 2022)",
     ylab="Return",ylim=c(-0.035,0.04))
lines(Date, MaxRet, col="blue", lty=2)
lines(Date, DollNeut, col="green", lty=2)
grid(nx=20,ny=20)

# Add a legend
legend("topleft", legend=c("MinVar", "MaxRet","DollNeut"),
       col=c("red", "blue","green"),
       lty=1, cex=0.8)
```

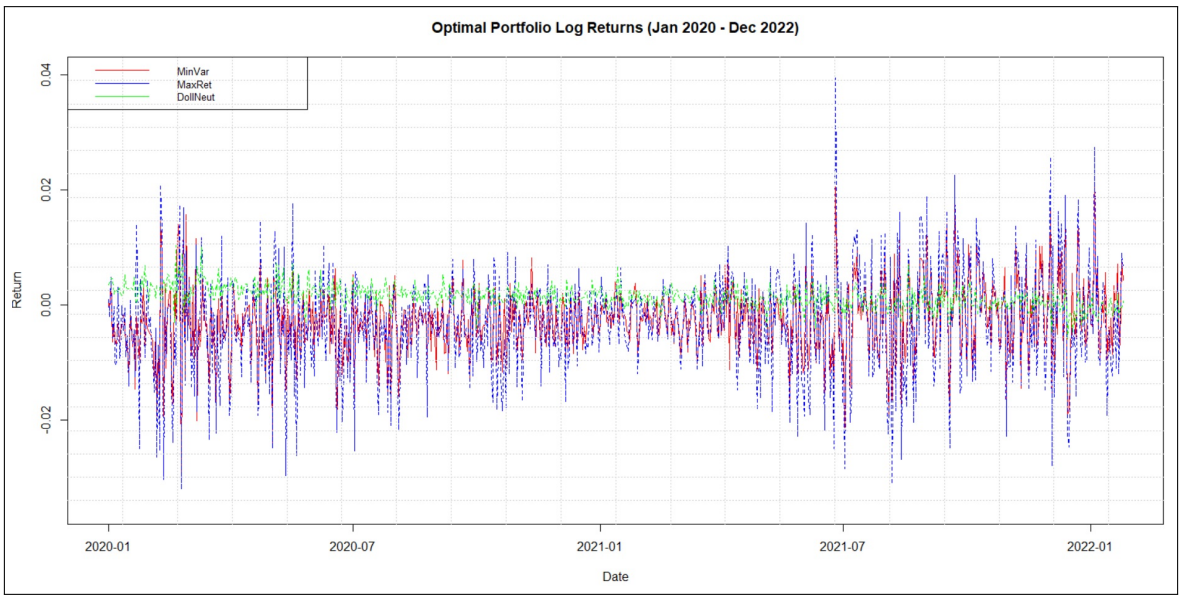

Figure 9: Optimal Portfolio Log Returns.

```r
#######################
# Kernel Density Plot.
#######################
windows(8,10)
plot(density(MinVar), col = "red",
     main="Densities (Optimal Portfolio Returns)",
     lwd=1.5)
lines(density(MaxRet), col = "blue",lwd=1.5)
lines(density(DollNeut), col = "green",lwd=1.5)

# Add legend to density plot
legend("topright",
       legend = c("Min-Var", "Max-Ret", "DollNeut"),
       col = c("red", "blue", "green"),
       lwd=1.5, lty = 1)
```

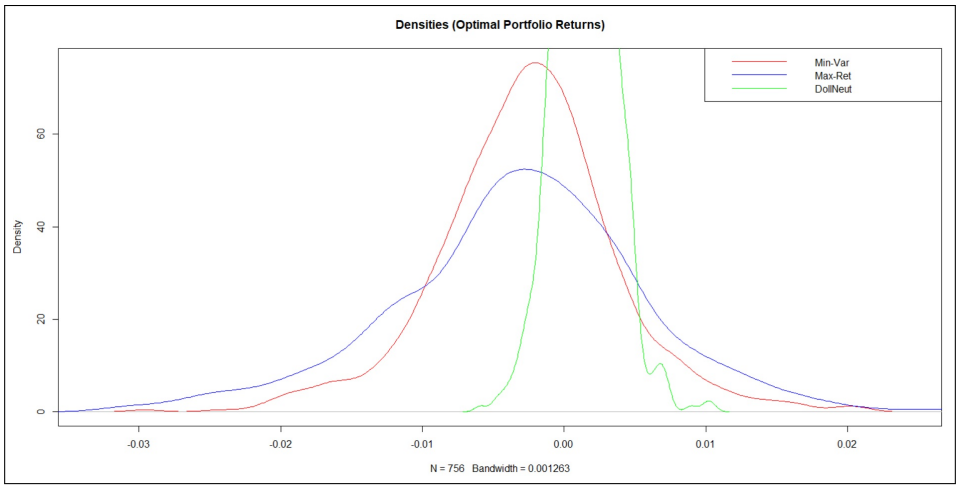

Figure 10: Kernel Density (Optimal Portfolio Returns).

```
################################
# Kernel Density Plot [Polygon].
################################
windows(8,10)
plot(density(MinVar), col = "red",
     main="Densities (Optimal Portfolio Returns)",
     lwd=1.5)
polygon(density(MinVar), col = "red")
lines(density(MaxRet), col = "blue",lwd=1.5)
polygon(density(MaxRet), col = "blue")
lines(density(DollNeut), col = "green",lwd=1.5)
polygon(density(DollNeut), col = "green")

# Add legend to density plot
legend("topright",                                  # Add legend to density
       legend = c("Min-Var", "Max-Ret", "DollNeut"),
       col = c("red", "blue", "green"),
       lwd=1.5, lty = 1)
```

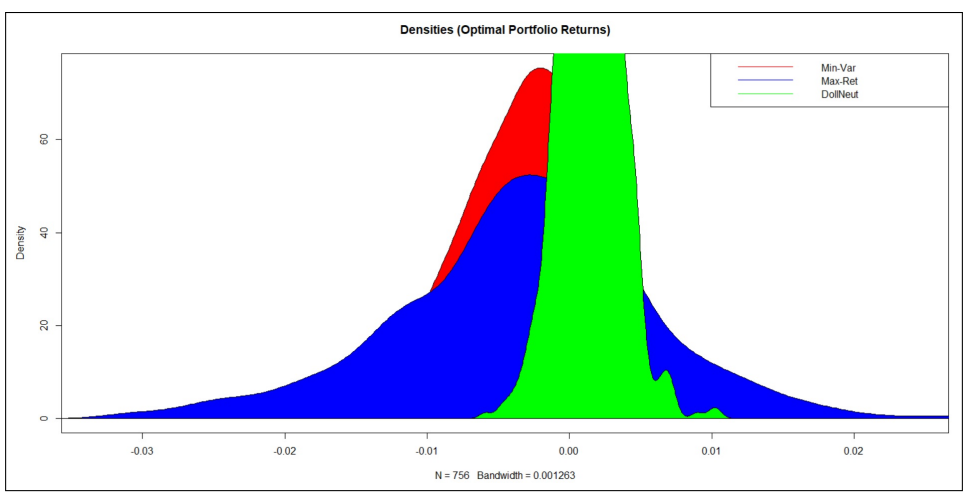

Figure 11: Kernel Density (Optimal Portfolio Returns).

```r
##########################################
# APPENDIX 1: Different Utility Functions.
##########################################
x=sigma=seq(0,5,by=0.1)

Ulin=x
windows(8,10)
plot(x,Ulin,type="l",col="red",
     main="Liner Utility Function.")
```

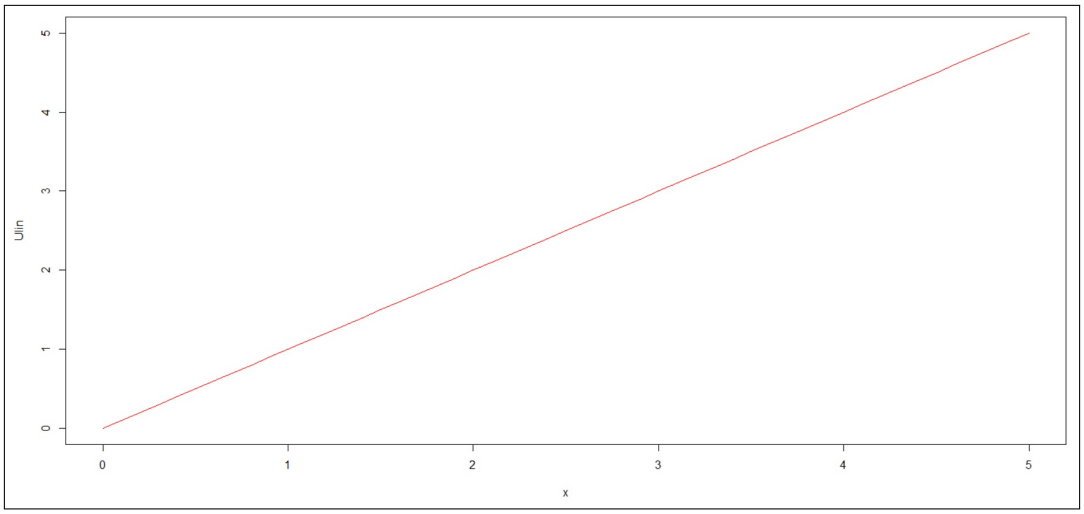

Figure 12: Linear Utility Function.

```r
a=3
e=2.71828
Uexp = -e^(-a*x)
plot(x,Uexp,type="l",col="blue",
     main="Exponential Utility Function.")
```

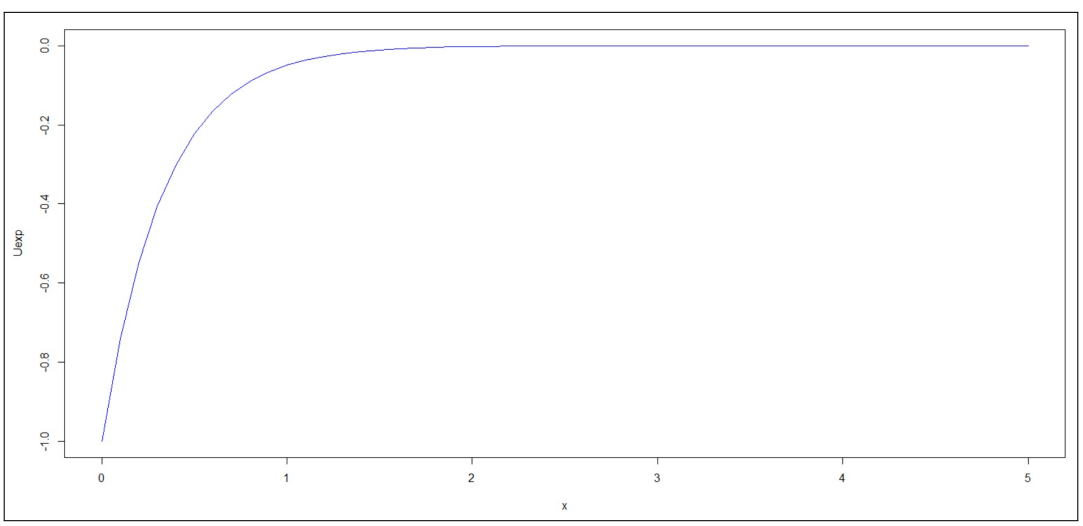

Figure 13: Exponential Utility Function.

```r
Ulog = log(x)
plot(x,Ulog,type="l",col="orange",
     main="Logarithmic Utility Function.")
```

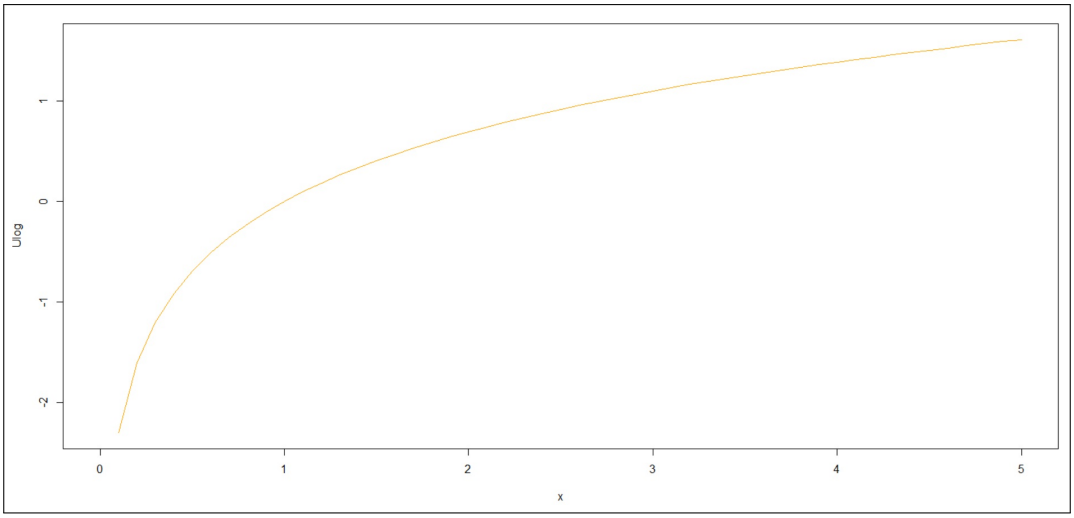

Figure 15: Logarithmic Utility Function.

```r
b=0.5
Upow = b*x^(b)
plot(x,Upow,type="l",col="green",
     main="Power Utility Function.")
```

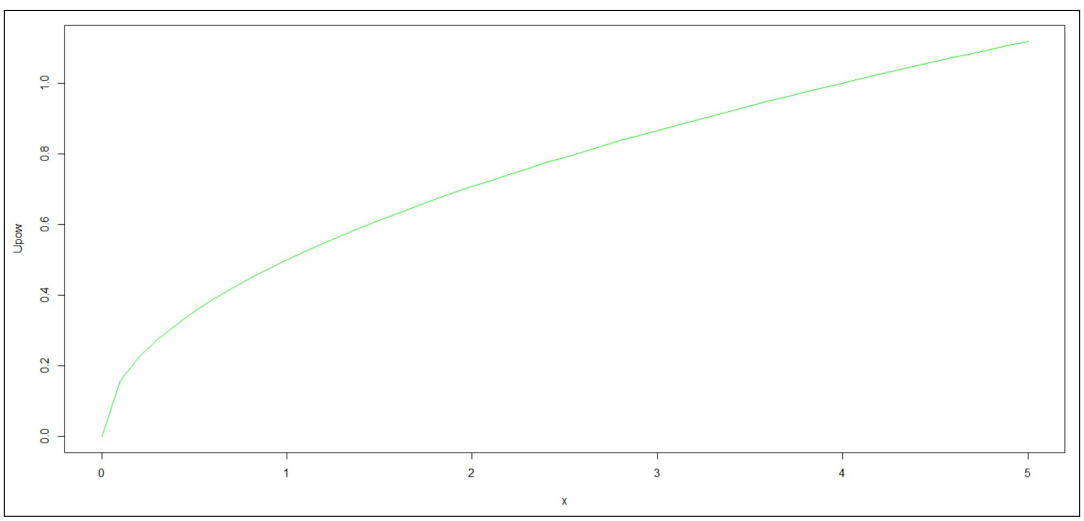

Figure 15: Power Utility Function.

```r
b=0.15
Uquad = x - b*x^2
plot(x,Uquad,type="l",col="purple",
     main="Quadratic Utility Function.")
```

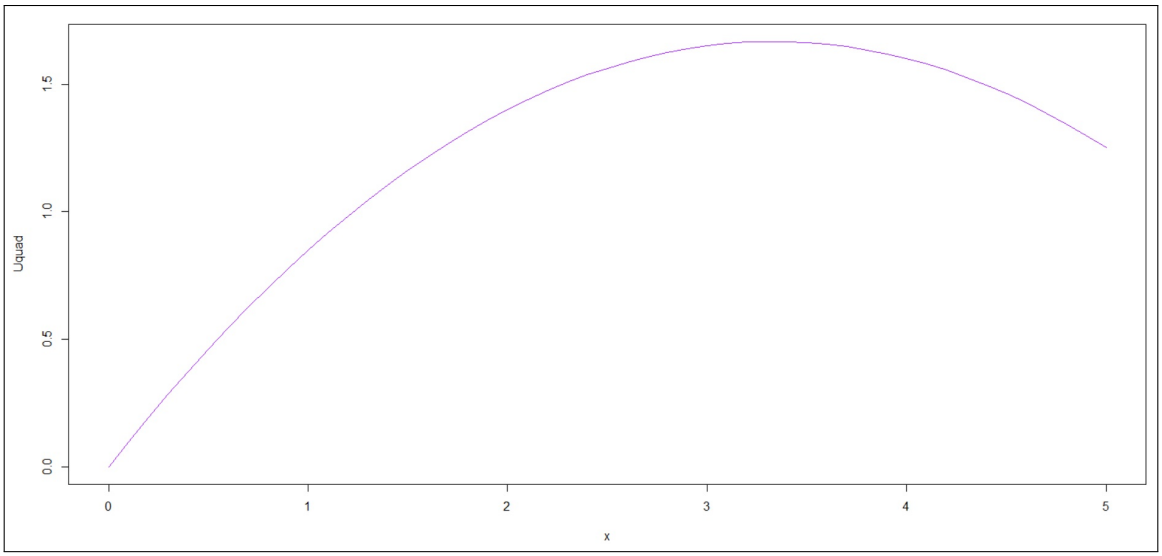

Figure 16: Quadratic Utility Function.

```
#######################################
# Plot all Utility Functions Together.
#######################################
windows(8,10)
plot(x,Ulin, col = "red",
     main="Different Utility Functions [U(x)]",
     lwd=2.5,type="l",ylim=c(-3,3),xlim=c(0,5),
     ylab="Utility")
lines(x,Uexp, col = "blue",lwd=2.5,type="l")
lines(x,Ulog, col = "orange",lwd=2.5,type="l")
lines(x,Upow, col = "green",lwd=2.5,type="l")
lines(x,Uquad, col = "purple",lwd=2.5,type="l")
# Add axis lines
abline(v = 0)
abline(h = 0)

# Add text (Utility Specifications)
text(3.25,3,"U(x) = x",col="red")
text(0.15,-0.25,"U(x) = -e^(ax)",col="blue")
text(4.75,1.75,"U(x) = log(x)",col="orange")
text(4.75,0.95,"U(x) = bx^b",col="green")
text(2.75,1.75,"U(x) = x - bx^2",col="purple")

# Add legend
legend("bottomright",                                   # Add legend to density
       legend = c("Linear U(x)", "Exponential U(x), (a = 3)",
                  "Logarithmic U(x)",
                  "Power U(x), (b = 0.5)",
                  "Quadratic U(x), (b = 0.15)"),
       col = c("red", "blue", "orange","green","purple"),
       lwd=1.5, lty = 1,cex = 0.75)
grid(nx=10,ny=10)
```

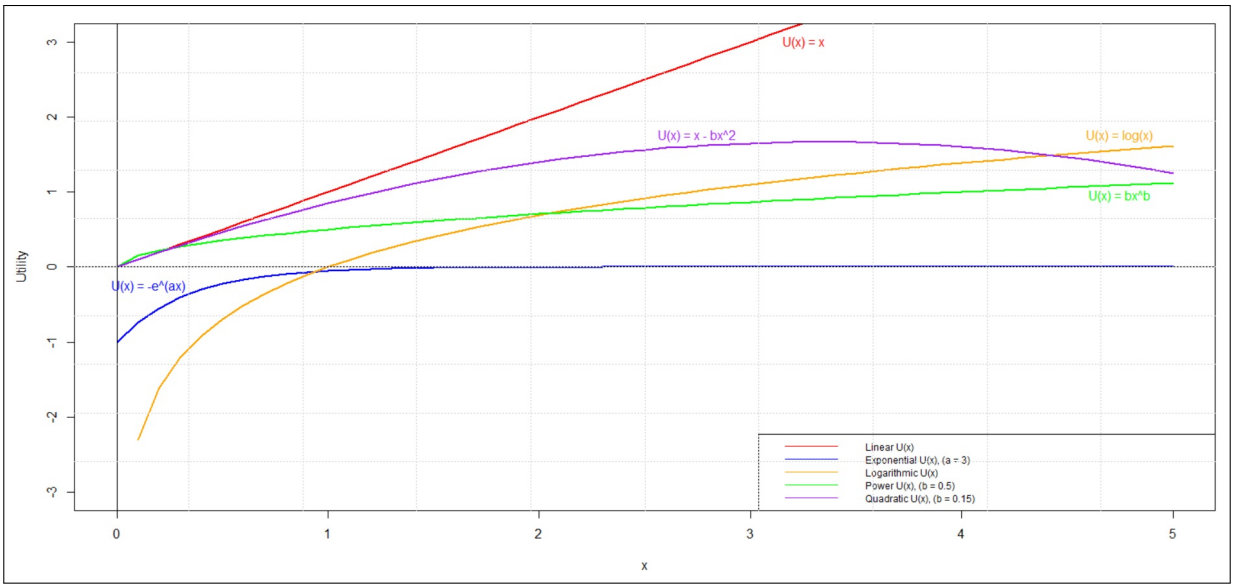

Figure 17: Different Utility Function $[U(x)]$.

```
##############################################
# APPENDIX 2: Portfolio Indifference Curves.
##############################################

# Initial Setting
#################
# Utility = 5
# Gamma = (1,8)
# Variance = (0,10)
###################

################################
# Portfolio Indifference Curves.
################################
U=5
sigma=seq(0,10,by=0.1)

##########################################
## Expected returns for different gammas.
##########################################

gamma1=1
Er1= U + 0.5*(sigma^2)*gamma1

gamma2=2
Er2= U + 0.5*(sigma^2)*gamma2

gamma3=3
Er3= U + 0.5*(sigma^2)*gamma3


gamma4=4
Er4= U + 0.5*(sigma^2)*gamma4
```

```r
gamma5=5
Er5= U + 0.5*(sigma^2)*gamma5

gamma6=6
Er6= U + 0.5*(sigma^2)*gamma6

gamma7=7
Er7= U + 0.5*(sigma^2)*gamma7

gamma8=8
Er8= U + 0.5*(sigma^2)*gamma8

#########################################
# Plotting Portfolio Indifference Curves.
#########################################
windows(8,10)
plot(sigma,Er1,type="l",col="red",lwd=3,
     main="Utility: U = Er + 1/2*(sigma^2)*(gamma).",
     xlab = "Std. Deviation (%)",ylab = "Expected Return (%)")
lines(sigma,Er2,type="l",col="blue",lwd=3)
lines(sigma,Er3,type="l",col="orange",lwd=3)
lines(sigma,Er4,type="l",col="green",lwd=3)
lines(sigma,Er5,type="l",col="yellow",lwd=3)
lines(sigma,Er6,type="l",col="violet",lwd=3)
lines(sigma,Er7,type="l",col="gray",lwd=3)
lines(sigma,Er7,type="l",col="black",lwd=3)
legend("bottomright", legend=c("gamma = 1", "gamma = 2","gamma = 3",
                               "gamma = 4","gamma = 5","gamma = 6",
                               "gamma = 7","gamma = 8"),
       col=c("red", "blue","orange","green","yellow",
             "violet","gray","black"),
       lty=1, cex=1.2)
grid(nx=20,ny=20)
```

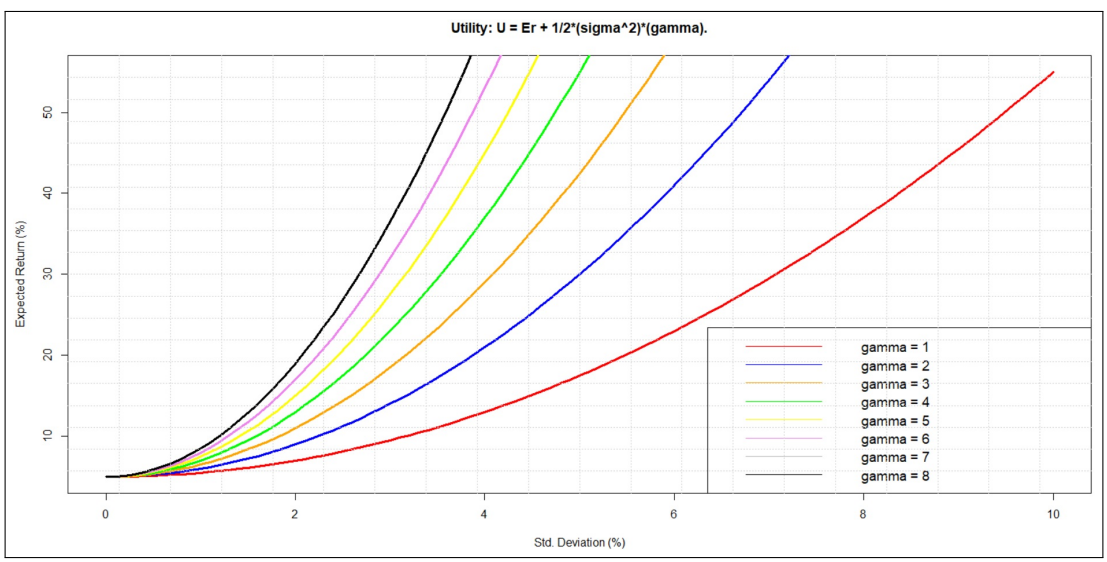

Figure 18: Portfolio Indifference Curves.

```
#####################
# END OF THE LECTURE.
#####################
```

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Spring 2026.